1

Charges and Fields

1.1. Basic Concepts

Electrostatics deals with the behaviour of stationary electric charges. There are two kinds of electric charges : *positive* and *negative*. Like charges repel each other and unlike charges attract each other.

All experiments so far have shown that electric charge in nature comes in units of one magnitude only. This magnitude is denoted by e and is called the electronic charge. All charges in nature occur in integral multiples of this basic unit [redacted] where n is either a positive or negative integer. Thus we say that charge exists in discrete packets rather than in continuous amounts. The charge is, therefore, *quantised.*

The *law of conservation of electric charge* states that the algebraic sum of the electric charges remains constant in a closed system. This implies that *charge can neither be created nor destroyed.*

[redacted] charges.

So far as their electrostatic behaviour is concerned, materials are divided into two categories : [redacted]). Bodies which allow the charge or electricity to pass through them are called conductors, *e.g.*, metals, human body, earth, graphite, charcoal etc. Bodies which do not allow the charge or electricity to pass through them are called insulators, *e.g.*, glass, mica, ebonite, plastics etc.

[redacted]

[redacted]. *The force between two point charges is directly proportional to the product of the charges, and inversely proportional to the square of the distance between them.*

The direction of the force will be along the line joining the two charges. The force is attractive if the two charges are of opposite nature and repulsive if the two charges are of the same type. The force depends on the nature of the medium in which the given two charges are situated.

Let q_1 and q_2 be two point charges situated at A and B separated by a distance r (Fig. 1.1). $\hat{r}$ is the unit vector in the direction from A to B. Then the force ($\mathbf{F}_{12}$) exerted by charge q_1 on charge q_2 is

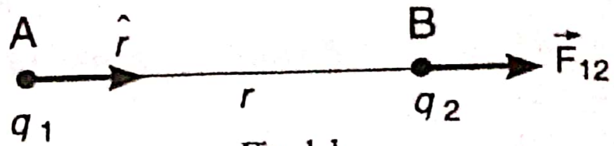


Fig. 1.1

$$\mathbf{F}_{12} = C \frac{q_1 q_2}{r^2} \hat{r} \qquad ...(1)$$

where C is a constant. In SI units, $C = 1/(4\pi\varepsilon_0)$ when the charges are situated in vacuum. Here ε_0 is called the *permittivity of free space* (*i.e.*, vacuum).

The measured value of ε_0 is

$\varepsilon_0 = 8.85418 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$ (or $\mathrm{Fm^{-1}}$).

This gives $1/(4\pi\varepsilon_0) = 9 \times 10^9\ \mathrm{Nm^2\ C^{-2}}$.

When there is vacuum between the two charges,

Coulomb's law can be written as

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r^2} \hat{r} \qquad ...(2)$$

Force exerted by q_2 on q_1 is equal and opposite to that exerted on q_2 by q_1 so that $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

If the charges are situated in a medium of *permittivity* ε, then the force between the charges is

$$\mathbf{F}_{12\,(m)} = \frac{1}{4\pi\varepsilon} \frac{q_1 q_2}{r^2} \hat{r} \qquad ...(3)$$

Dividing Eq. (2) by Eq. (3), we get

$$\frac{\mathbf{F}_{12}}{\mathbf{F}_{12\,(m)}} = \frac{\varepsilon}{\varepsilon_0} = \varepsilon_r \qquad ...(4)$$

ε_r is called the *relative permittivity* of the medium. It is defined as *the ratio of the permittivity of the medium to that of free space*. (An older name for ε_r is the *dielectric constant* of the material). ε_r is a dimensionless quantity. The value of ε_r for air is 1. To distinguish ε from ε_r we call the former the *absolute permittivity* of the material. The units of ε are $\mathrm{Fm^{-1}}$.

In Eq. (2), if $q_1 = q_2 = 1$ and $r = 1$, we have

$$F = \left(\frac{1}{4\pi\varepsilon_0}\right) \frac{q_1 q_2}{r^2} = (9 \times 10^9) \frac{1 \times 1}{1^2} = 9 \times 10^9\ \mathrm{N}.$$

The SI unit of charge is the *coulomb*.

A coulomb is defined as the quantity of charge which, when placed at a distance of 1 metre in air or vacuum from an equal and similar charge, *experiences a repulsive force of 9×10^9 N.*

A coulomb is also defined as the amount of charge that passes through any cross-section of a wire in 1 second if there is a current of 1 ampere in the wire.

Note. Throughout the remaining portion of the book, we shall always assume the medium to be air unless otherwise stated.

1.3. Superposition Principle

If we have n point charges, they act independently in pairs. The force on any one of them, let us say q_1, is given by the vector sum

$$\mathbf{F}_1 = \mathbf{F}_{12} + \mathbf{F}_{13} + \mathbf{F}_{14} + \mathbf{F}_{15} + \ldots + \mathbf{F}_{1n} \qquad \ldots(5)$$

Here, $\mathbf{F}_1$ represents the force on charge q_1 by all the charges. $\mathbf{F}_{12}$ is the force acting on particle 1 owing to the presence of particle 2 and so on.

[redacted] it. We speak of an *electric field* in this region. We represent it by a vector **E** defined by

[redacted] ...(6)

[redacted]

The direction of the vector **E** is that of the vector **F**. The SI unit for the electric field is the newton/coulomb (NC^{-1}) or volt/metre ($V\,m^{-1}$).

Several other terms are commonly used, for example, electric field strength, electric field intensity or intensity of the electric field. The words "strength" or "intensity" are really unnecessary.

Eq. (6) may be written as

$$\mathbf{F} = q\,\mathbf{E} \qquad \ldots(7)$$

i.e., the force **F** exerted on a charge q at a point where the electric field is **E** equals the product of the electric field and the charge.

[redacted]e. Thus

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \ldots = \sum \mathbf{E}_i$$

$$= \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{r_1^2}\hat{r}_1 + \frac{q_2}{r_2^2}\hat{r}_2 + \frac{q_3}{r_3^2}\hat{r}_3 + \ldots\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i^2}\hat{r}_i \qquad \ldots(8)$$

For a continuous charge distribution, the electric field **E** at any point is given by

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2}\hat{r} \qquad \text{...(9)}$$

[illegible]

Let P be a point lying in vacuum at a distance **r** from a point charge q lying at O (Fig. 1.2). Let a test charge q_0 be placed at P. According to Coulomb's law, the force **F** acting on q_0 due to q is

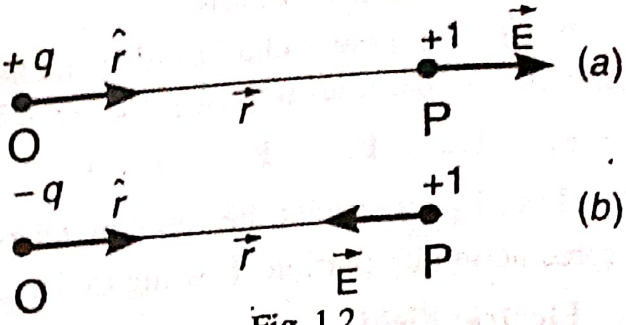


Fig. 1.2

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\hat{r}$$

The electric field at the point P is, by definition, given by the force per unit test charge.

$$\therefore \quad \mathbf{E} = \frac{\mathbf{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \qquad \text{...(10)}$$

The direction of **E** is along the line joining O and P, pointing outward if q is positive [Fig. 1.2 (a)] and inward if q is negative [Fig. 1.2 (b)].

[illegible] *A positive charge of $q_1 = 2\times10^{-7}$ C is placed at a distance of 0.15m from another positive charge of $q_2 = 8\times10^{-7}$ C. At what point on the line joining them is the electric field zero?*

Sol. The point P at which the electric field is zero must lie *between* $+q_1$ and $+q_2$ because only here the forces exerted by $+q_1$ and $+q_2$ on a test charge at P will be oppositely directed. Let x be the distance of P from q_1 (Fig. 1.3).

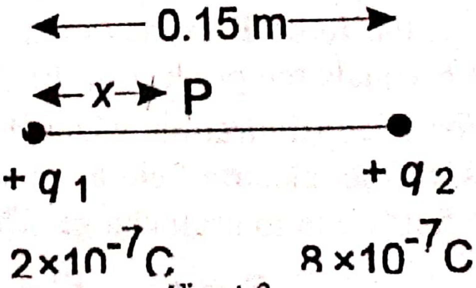


Fig. 1.3

Let E_1 and E_2 be the magnitudes of electric fields at P due to q_1 and q_2 respectively. Then

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{x^2} \quad \text{and} \quad E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{(0.15-x)^2}$$

E_1 is directed toward q_2 and E_2 toward q_1.

Since the electric field at P is zero, E_1 and E_2 are equal and opposite.

$$E_1 = E_2$$

$$\frac{1}{4\pi\varepsilon_0}\frac{q_1}{x^2} = \frac{1}{4\pi\varepsilon_0}\frac{q_2}{(0.15-x)^2}$$

$$(0.15 - x)^2 = (q_2/q_1)x^2$$

$$(0.15 - x)^2 = 4x^2 \qquad (\because q_2/q_1 = 4)$$

$$3x^2 + 0.3x - 0.0225 = 0$$

Solving, $\quad x = -0.15;\ 0.05.$

P must lie *between* $+q_1$ and $+q_2$. So the first value is discarded. Hence the electric field is zero at a distance of 0.05 m from the charge q_1.

[redacted] 2. *ABCD is a square of 4 cm side. Charges of 16×10^{-9}, -16×10^{-9} and 32×10^{-9} C are placed at the points A, C and D respectively. Find the intensity of the electric field at point B.*

Sol. Let charges at A, C and D produce electric fields E_A, E_C and E_D at B (Fig. 1.4).

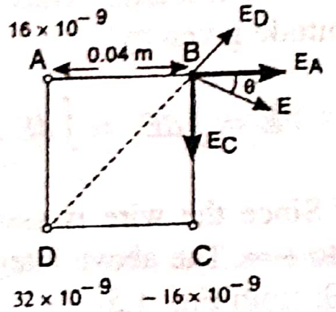


Fig. 1.4

$$E_A = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

$$= (9\times10^9)\times\frac{16\times10^{-9}}{(0.04)^2}$$

$$= 9\times10^4 \text{ along } AB.$$

$$E_C = (9\times10^9)\times\frac{(-16\times10^{-9})}{(0.04)^2} = 9\times10^4 \text{ along } BC.$$

$$E_D = (9\times10^9)\times\frac{32\times10^{-9}}{(\sqrt{32}\times10^{-4})^2} = 9\times10^4 \text{ along } DB.$$

$\therefore$ the resultant electric field at B,

$$\mathbf{E} = \mathbf{E}_A + \mathbf{E}_C + \mathbf{E}_D$$

$$\therefore\quad E = \sqrt{E_x^2 + E_y^2} = \sqrt{(E_A + E_D\cos45^\circ)^2 + (E_C - E_D\sin45^\circ)^2}$$

$$= \sqrt{E_A^2 + E_D^2 + E_C^2 + 2E_D(E_A\cos45^\circ - E_C\sin45^\circ)}$$

$$= 9\sqrt{3}\times10^4 \text{ NC}^{-1}$$

$$\tan\theta = \frac{E_y}{E_x} = \frac{E_C - E_D\sin45^\circ}{E_A + E_D\cos45^\circ} = \frac{(1 - 1/\sqrt{2})\,9\times10^4}{(1 + 1/\sqrt{2})\,9\times10^4} = 0.171$$

$\therefore\ \theta = 9^\circ 44'$.

Example 3. *Calculate the electric field at a distance y from an infinite line charge having linear density of charge* λ.

Sol. AB is an infinite line charge having linear charge density λ and placed in vacuum (Fig. 1.5). P is a point at a distance y from O. Consider an element of length dx.

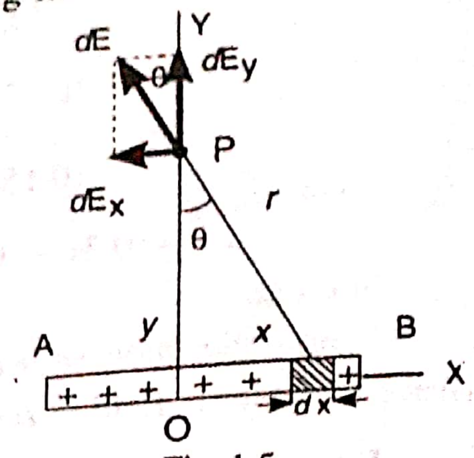


Fig. 1.5

Magnitude of the electric field at P due to this element $= dE = \dfrac{1}{4\pi\varepsilon_0}\dfrac{\lambda\, dx}{r^2}$

The vector $d\mathbf{E}$ has two components

(i) $dE_x = -\, dE \sin\theta$ and

(ii) $dE_y = dE \cos\theta$

$\int dE_x$ is zero because for every charge element in the + ve direction of x axis, there is a corresponding element in the – ve direction of x axis. All the y components point along the same direction OP and simply add up to give the resultant. Thus the resultant field points along OP and has a magnitude given by

$$E = \int dE_y = \int dE\cos\theta = \int_{-\infty}^{\infty} \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dx}{(y^2 + x^2)}\cos\theta$$

Since the wire is assumed to be *infinite*, the limits for x are from $-\infty$ to $+\infty$. The above integral is evaluated by changing the variable from x to θ; from Fig. 1.5,

$$x = y\tan\theta$$

$$\therefore\ dx = y\sec^2\theta\, d\theta$$

When $x = -\infty$, $\theta = -\pi/2$; When $x = +\infty$, $\theta = +\pi/2$.

$$E = \frac{\lambda}{4\pi\varepsilon_0}\int_{-\pi/2}^{+\pi/2}\frac{y\sec^2\theta\, d\theta}{y^2(1+\tan^2\theta)}\cos\theta = \frac{\lambda}{4\pi\varepsilon_0 y}\int_{-\pi/2}^{+\pi/2}\cos\theta\, d\theta$$

$$\therefore\ E = \frac{\lambda}{2\pi\varepsilon_0 y}$$

The direction of $\mathbf{E}$ is along OP if the line charge is positive and along PO if negative.

1.6. [illegible].

Consider two charges $-q$ at point A and $+q$ at point B, the distance between them being $2d$ (Fig. 1.6). Such a charge configuration is called

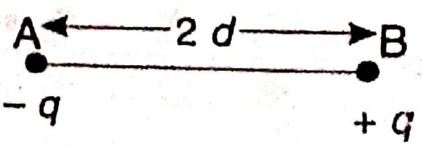


Fig. 1.6

an *electric dipole.* The magnitude of the dipole moment **p** is given by the p [illegible] m.

$$p = q \times 2d \qquad ...(11)$$

Direction of **p** is from $-q$ to $+q$.

The units of **p** are Cm.

1.7. [illegible]

Consider a dipole of moment $p = 2dq$ placed in a uniform electric field **E** · **p** makes an angle θ with the field direction (Fig. 1.7). The charge $+q$ experiences a force $q\mathbf{E}$ in the direction of the field. The charge $-q$ experiences an equal force in the opposite direction. Thus the net linear force on the dipole is zero. The two forces are not collinear. So they form a clockwise torque which tends to set the dipole in the direction of the electric field. T[illegible]e [illegible]s

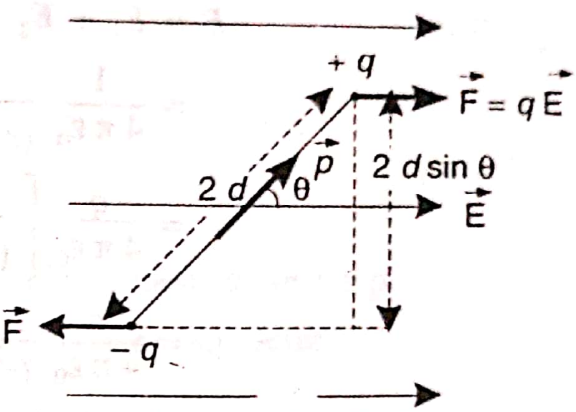


Fig. 1.7

$$\tau = qE\,(2d \sin\theta)$$

$$= pE \sin\theta \qquad (\because\ q \times 2d = p)$$

The torque can be written in vector form as

$$\vec{\tau} = \mathbf{p} \times \mathbf{E} \qquad ...(12)$$

Suppose we choose the potential energy of a dipole in an external field to be zero when $\theta = 90^\circ$. Let W be the work done *by an external agent* to turn the dipole from its reference orientation to angle θ. Then

$$W = \int_{90^\circ}^{\theta} \tau\, d\theta = \int_{90^\circ}^{\theta} pE \sin\theta\, d\theta = -pE\cos\theta$$

This work done is stored as potential energy in the dipole.

$$\therefore\ U(\theta) = -pE\cos\theta = -\mathbf{p} \cdot \mathbf{E} \qquad ...(13)$$

1.8. [illegible]t

Fig. 1.8. shows two charges of magnitude q but of opposite sign, separated by a distance $2d$.

We want to calculate the electric field due to the dipole at a point P along its axis at a distance r from the centre O of the dipole. Let the medium be air or vacuum.

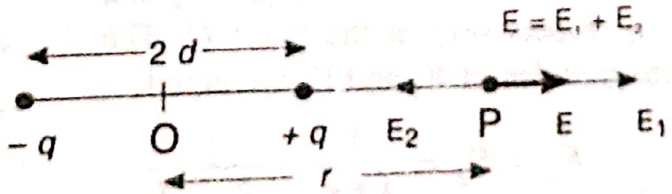


Fig. 1.8

The electric field at P due to the positive charge is given by

$$E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-d)^2}$$

The electric field at P due to the negative charge

$$E_2 = \frac{1}{4\pi\varepsilon_0}\frac{(-q)}{(r+d)^2}$$

E_1 and E_2 act in opposite directions and $E_1 > E_2$.

Therefore, the magnitude of the electric field at P is

$$E = E_1 + E_2$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-d)^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+d)^2}$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{(r-d)^2} - \frac{1}{(r+d)^2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{4rd}{(r^2-d^2)^2}$$

If P is far from the dipole, then $r >> d$.

$$\therefore \quad E = \frac{q}{4\pi\varepsilon_0}\frac{4rd}{r^4} = \frac{q}{4\pi\varepsilon_0}\frac{4d}{r^3}$$

But $q \times 2d = p =$ *Electric dipole moment* of the dipole.

$$\therefore \quad E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3} \qquad ...(14)$$

The direction of **E** is in the direction of the dipole moment vector **p**.

1.9. E [redacted] e

The perpendicular bisector of the line joining the charges is the equatorial line. Q is a point at a distance r from the centre O of the dipole on the equatorial line (Fig. 1.9).

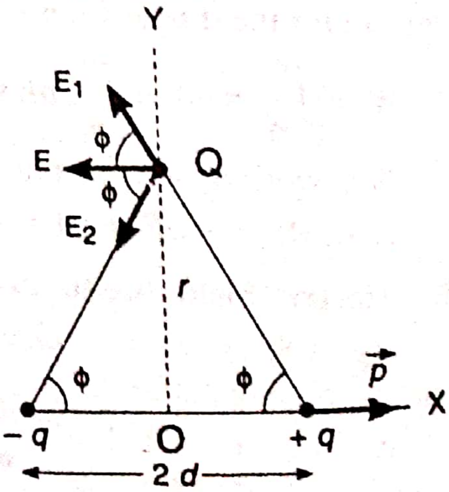


Fig. 1.9

Let $\mathbf{E_1}$ and $\mathbf{E_2}$ denote the electric fields due to charges $+q$ and $-q$ respectively at the point Q. The magnitudes of $\mathbf{E_1}$ and $\mathbf{E_2}$ are equal.

$$E_1 = E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(d^2+r^2)}$$

The directions of $\mathbf{E_1}$ and $\mathbf{E_2}$ are as shown in Fig. 1.9

Now we resolve $\mathbf{E_1}$ and $\mathbf{E_2}$ into components along the X and Y directions. The Y components of $\mathbf{E_1}$ and $\mathbf{E_2}$ cancel out. But the X

components of $\mathbf{E}_1$ and $\mathbf{E}_2$ add to yield the resultant field $\mathbf{E}$. Thus

$$E = E_1 \cos\phi + E_2 \cos\phi = 2E_1 \cos\phi \qquad (\because E_1 = E_2)$$

But $$\cos\phi = \frac{d}{(d^2 + r^2)^{1/2}}$$

$$\therefore \quad E = 2 \cdot \frac{1}{4\pi\varepsilon_0} \frac{q}{(d^2 + r^2)} \cdot \frac{d}{(d^2 + r^2)^{1/2}} = \frac{q\,(2d)}{4\pi\varepsilon_0 (d^2 + r^2)^{3/2}}$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{p}{(d^2 + r^2)^{3/2}} \qquad \text{...(15)}$$

d is negligible compared to r.

$$\therefore \quad E = \frac{1}{4\pi\varepsilon_0} \frac{p}{r^3} \qquad \text{...(16)}$$

Direction of $\mathbf{E}$ is opposite to that of dipole moment vector $\mathbf{p}$.

1.10. [illegible]

Consider a point R having polar coordinates r and θ (Fig. 1.10). O is the centre of the dipole. Resolve the dipole moment vector $\mathbf{p}$ into two components, one parallel and the other perpendicular to OR. The point R is on the axis of a dipole of moment $p\cos\theta$ and on the equatorial line of a dipole of moment $p\sin\theta$.

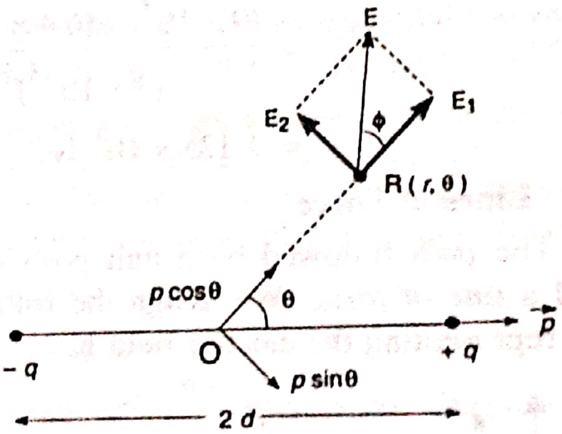


Fig. 1.10

$$\left.\begin{array}{l}\text{Field due to the}\\ \text{component } p\cos\theta\end{array}\right\} = E_1 = \frac{2p\cos\theta}{4\pi\varepsilon_0 r^3}$$

$$\left.\begin{array}{l}\text{Field due to the}\\ \text{component } p\sin\theta\end{array}\right\} = E_2 = \frac{p\sin\theta}{4\pi\varepsilon_0 r^3}$$

$\therefore$ Magnitude of the resultant field

$$E = \sqrt{E_1^2 + E_2^2} = \frac{p}{4\pi\varepsilon_0 r^3} \sqrt{(4\cos^2\theta + \sin^2\theta)}$$

$$= \frac{p}{4\pi\varepsilon_0 r^3} \sqrt{1 + 3\cos^2\theta} \qquad \text{...(17)}$$

The resultant field $\mathbf{E}$ makes an angle ϕ with OR.

$$\tan\phi = \frac{p\sin\theta/(4\pi\varepsilon_0 r^3)}{2p\cos\theta/(4\pi\varepsilon_0 r^3)} = \frac{1}{2}\tan\theta \qquad ...(18)$$

Eq. (17) gives the magnitude and Eq. (18) gives the direction of the electric field.

[redacted]. *A dipole is consisting of an electron and proton, 4×10^{-10} m apart. Compute the electric field at a distance of 2×10^{-8} m on a line making an angle of 45^o with the dipole axis from the centre of the dipole.*

Sol. Dipole moment $p = q\times 2d$

$$= (1.6\times10^{-19})\times(4\times10^{-10})$$

$$= 6.4\times10^{-29}\ \text{Cm.}$$

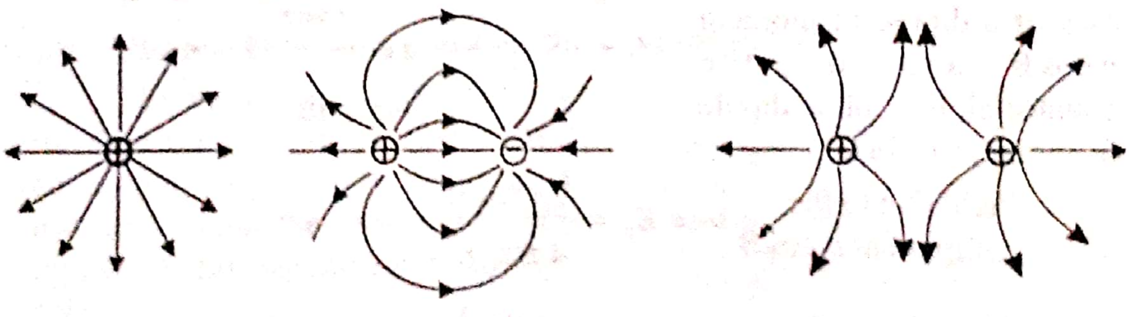
Electric field $E = \dfrac{p}{4\pi\varepsilon_0 r^3}\sqrt{1+3\cos^2\theta}$

$$= \frac{(9\times10^9)\times(6.4\times10^{-29})}{(2\times10^{-8})^3}\sqrt{1+\cos^2 45^o}$$

$$= 1.138\times10^5\ \text{NC}^{-1}.$$

1.11. [redacted]

The path followed by a unit positive charge in an electric field is called a *line of force*. We assign the following properties to the lines of force representing the electric field **E**.



(a) (b) (c)

Fig. 1.11

1. Lines of force start from positive charges and terminate on negative charges. Fig. 1.11 shows the lines of force for some simple arrangements of point charges.

2. Lines of force never intersect.

3. The tangent to a line of force at any point gives the direction of the electric field **E** at that point.

4. The lines of force are drawn so that the number of lines per unit area, in a plane at right angles to the lines, is proportional to the magnitude

of **E**. This means that where the lines of force are close together, E is large and that where they are far apart, E is small.

5. It is assumed that *each unit positive charge gives rise to* $1/\varepsilon_0$ *lines of force in free space.*

EXERCISE I

1. Find the potential energy of an electric dipole placed in a uniform external electric field.
2. Derive expressions for the electric field at a point on the (*a*) axial line (*b*) equatorial line, due to an electric dipole.
3. Explain electric dipole and electric dipole moment. Derive an expression for the electric field at any point due to an electric dipole.
4. Two pith balls, each weighing 100 mg and suspended from the same point by silk threads 20 cm long, are equally charged and repel each other to a distance of 10 cm. Calculate the charge on each ball.
   [Ans. $q = 1.676 \times 10^{-8}$ C]
5. Three small balls each of mass 10 gm are suspended separately from a common point by silk threads, each one metre long. The balls are identically charged and hung at the corners of an equilateral triangle of side 10 cm. What is the charge on each ball ? **[Ans.** 6.2×10^{-8} C]
6. Two similar balls of mass m are hung from silk threads of length l and carry similar charges q. The angle θ is the angle through which each string is deflected from the vertical. Assume that θ is so small that $\tan\theta$ can be replaced by $\sin\theta$. Show that the distance of separation between the balls is given by $x = \left[\dfrac{q^2 l}{2\pi\varepsilon_0 mg}\right]^{\frac{1}{3}}$.
7. A cube of edge a carries a point charge q at each corner. Calculate the resultant force on any one of the charges.
   [Ans. $\dfrac{0.261\, q^2}{\varepsilon_0 a^2}$ along the diagonal of the cube].
8. [redacted]

[Hint : $\tau_{max} = 2 \times 10^{-8} \times 1.5 \times 10^5 \times \sin 90° = 3 \times 10^{-3}$ Nm

Work done in rotating from θ_1 to $\theta_2 = pE(\cos\theta_1 - \cos\theta_2)$
$= 2 \times 10^{-8} \times 1.5 \times 10^5 (\cos 0 - \cos 180) = 6 \times 10^{-3}$ J

2

Gauss's Law and Its Applications

2.1. Flux of the Electric Field

Consider a closed surface S immersed in a nonuniform electric field (Fig. 2.1). Divide the surface into small squares of area dS. We represent each such element of area by a vector $d\mathbf{S}$, whose magnitude is the area dS. The direction of $d\mathbf{S}$ is defined to be that of the *outward-drawn normal* to the surface. Since the squares are very small, the electric field $\mathbf{E}$ for all points in the square is constant. The vectors $\mathbf{E}$ and $d\mathbf{S}$ make an angle θ with each other.

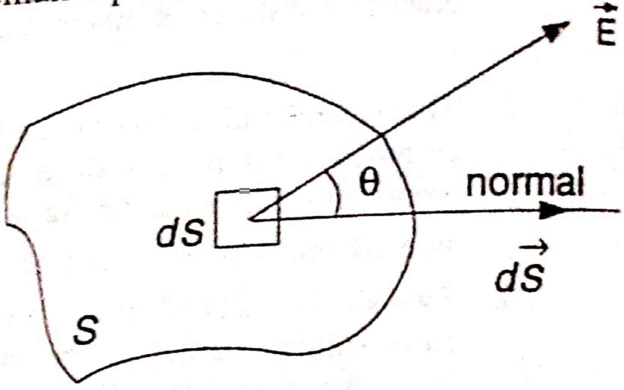


Fig. 2.1

The electric flux $d\phi$ through the area dS is defined as

$$d\phi = \mathbf{E} \cdot d\mathbf{S} = E\, dS \cos\theta \qquad ...(1)$$

$d\phi$ is $\gtrless 0$ depending on whether θ is $\lessgtr \frac{\pi}{2}$ respectively.

The total flux through the closed surface S is obtained by integrating the above equation over the surface. Thus

$$\phi = \oint d\phi = \oint \mathbf{E} \cdot d\mathbf{S} \qquad ...(2)$$

The circle on the integral sign indicates that the integration is to be taken over the entire (closed) surface. The flux of the electric field is a scalar. Its unit is Nm^2C^{-1} or Vm.

2.2. Gauss's Law

Statement. *The total flux of the electric field* $\mathbf{E}$ *over any closed surface is equal to* $1/\varepsilon_0$ *times the total net charge enclosed by the surface.*

$$\phi = \oint \mathbf{E} \cdot d\mathbf{S} = \frac{q}{\varepsilon_0} \qquad ...(1)$$

Explanation. This law relates the flux through any closed surface and the *net* charge enclosed within the surface. Here q is the net charge inside the closed surface. This closed hypothetical surface is called *Gaussian* surface. Gauss' law tells us that the flux of **E** through a closed surface S depends only on the value of the net charge inside the surface and not on the location of the charges. Charges outside the surface will not contribute to ϕ. Eq. (1) holds only when the charges are lying in vacuum or air.

Proof.

(i) For a charge inside the closed surface

Consider a single point charge $+q$ located at a point O inside a closed surface S (Fig. 2.2). Let dS be a small area element at a distance r from q.

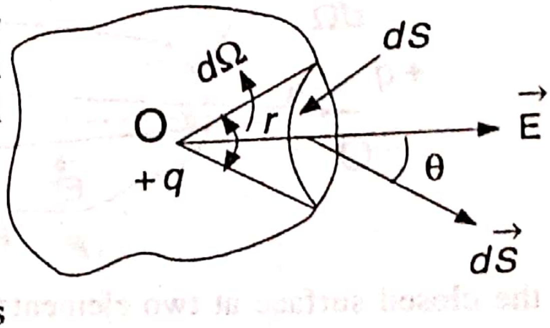


Fig. 2.2

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

The flux through the area dS is given by

$$d\phi = \mathbf{E}\,.\,d\mathbf{S}$$

$$= E\,dS\cos\theta$$

$$= \left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\right) dS\cos\theta$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{dS\cos\theta}{r^2}\right)$$

But $\dfrac{dS\cos\theta}{r^2} = d\Omega$ = Solid angle subtended by the area dS at O.

$$\therefore\ d\phi = \frac{q}{4\pi\varepsilon_0}\,d\Omega$$

The total flux through the entire closed surface S is given by

$$\phi = \oint d\phi = \frac{q}{4\pi\varepsilon_0}\oint d\Omega = \frac{q}{4\pi\varepsilon_0}\times 4\pi$$

($\because$ the total solid angle subtended by S at O is 4π, *i.e.*., $\oint d\Omega = 4\pi$).

$$\therefore \quad \phi = \frac{q}{\varepsilon_0}$$

Gauss' law holds even if there are a number of charges $q_1, q_2, \ldots q_n$ enclosed by a surface S (Fig. 2.3). We know from the principle of superposition that the electric field due to a number of charges is the vector sum of their *individual* fields.

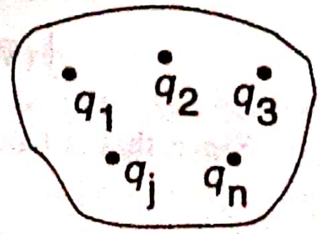


Fig. 2.3

$$\therefore \quad \phi = \oint \mathbf{E} \, . \, d\mathbf{S} = \frac{1}{\varepsilon_0} \sum_{i=1}^{n} q_i = \frac{Q}{\varepsilon_0}$$

Here, Q is the total charge inside the surface.

(ii) For a charge outside the closed surface

Consider a point charge $+ q$ situated at O outside the closed surface (Fig. 2.4). Let an elementary cone from O with small solid angle $d\,\Omega$ cut

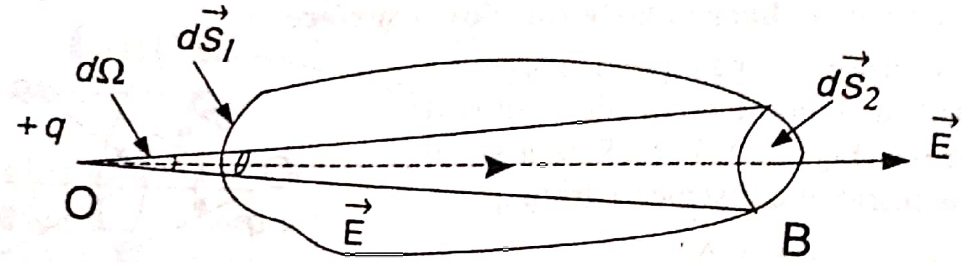


Fig. 2.4

the closed surface at two elements of area dS_1 and dS_2. Magnitude of flux through dS_1 and dS_2 are equal. Flux through dS_1 is an inward flux. Flux through dS_2 is an outward flux. Therefore,

$$\left.\begin{array}{c}\text{total flux through} \\ dS_1 \text{ and } dS_2\end{array}\right\} = \frac{-q}{4\pi\varepsilon_0} d\,\Omega + \frac{q}{4\pi\varepsilon_0} d\,\Omega = 0$$

The entire closed surface can be considered to be made of pairs of elements like dS_1 and dS_2. Thus the total flux, due to a charge *outside*, is zero.

2.3. Differential form of Gauss Law

Suppose the charge is distributed over a volume. Let ρ be the charge density. Then the total charge within the closed surface enclosing the volume is given by

$$Q = \int \rho \, dV \qquad \text{...(1)}$$

We can write integral form of Gauss law as

$$\oint \mathbf{E} \, . \, d\mathbf{S} = \frac{1}{\varepsilon_0} \int \rho \, dV \qquad \text{...(2)}$$

By Gauss divergence theorem,

$$\oint \mathbf{E} \, . \, d\mathbf{S} = \int (\nabla \, . \, \mathbf{E}) \, dV \qquad \text{...(3)}$$

Comparing Eq. (2) and Eq. (3),

$$\int (\nabla \, . \, \mathbf{E}) \, dV = \frac{1}{\varepsilon_0} \int \rho \, dV \qquad \text{...(4)}$$

Since this is true for *any* volume V, integrands must be equal.

$$\therefore \quad \nabla \, . \, \mathbf{E} = \frac{\rho}{\varepsilon_0} \qquad \text{...(5)}$$

2.4. An Insulated Conductor

Gauss's law can be used to make an important prediction, namely:

An excess charge, placed on an insulated conductor, resides entirely on its outer surface.

This hypothesis was shown to be true by experiment. Consider an insulated conductor carrying an excess charge q (Fig. 2.5). Consider a Gaussian surface (shown dotted) very close and *inside* the actual surface of the conductor. When an excess charge is placed at random on an insulated conductor, it will set up electric fields inside the conductor. These fields act on the charge carriers of the conductor and cause them to move. That is, they set up internal currents. These currents redistribute the excess charge in such a way that the internal electric fields are automatically reduced in magnitude. Finally the electric fields inside the conductor become zero everywhere, the currents automatically stop, and electrostatic conditions prevail. This redistribution of charge takes place in a negligible amount of time.

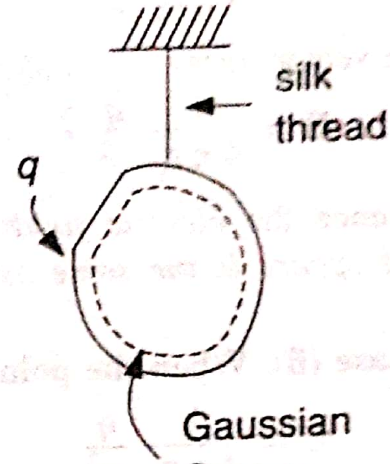


Fig. 2.5

If **E** is zero everywhere inside the conductor, it must be zero for every point on the Gaussian surface. Thus

$$\phi = \oint \mathbf{E} \, . \, d\mathbf{S} = 0$$

for the Gaussian surface. According to Gauss's law, this means that there must be no net charge inside the Gaussian surface. If the excess charge q is not *inside* this surface, it can only be *outside* it. That is, *the excess charge must be on the actual surface of the conductor.*

2.5. Electric Field due to a Uniformly Charged Sphere

A *spherically symmetric charge distribution* means the distribution of charge where the charge density ρ at any point depends only on the distance of the point from the centre and not on the direction. Consider a total charge q distributed uniformly throughout a sphere of radius R. Note that the sphere cannot be a conductor or, as we have seen, the excess charge will reside on its surface.

Case (i). When the point P lies outside the sphere

P is a point at a distance r from the centre O (Fig. 2.6). We have to find the electric field **E** at P. Draw a concentric sphere (shown dotted) of radius OP with centre O. This is the Gaussian surface. At all points of this sphere, the magnitude of the electric field is the same and its direction is perpendicular to the surface. Angle between **E** and dS is zero.

The flux through this surface is given by

$$\oint \mathbf{E} \cdot d\mathbf{S} = \oint E\, dS = E(4\pi r^2)$$

By Gauss's law,

$$E(4\pi r^2) = \frac{q}{\varepsilon_0}$$

or $$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

In vector form

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

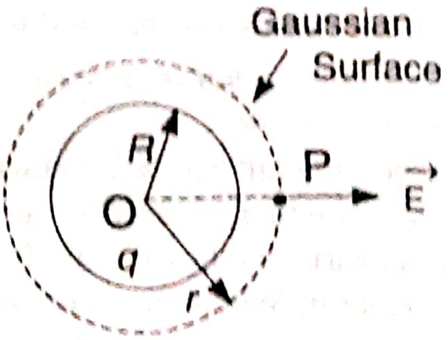


Fig. 2.6

Hence the *electric field at an external point due to a uniformly charged sphere is the same as if the total charge is concentrated at its centre.*

Case (ii). When the point lies on the surface. Here, $r = R$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}$$

Case (iii). When the point lies inside the sphere

P' is a point inside the sphere (Fig. 2.7). P' is at a distance r from the centre O. Draw a concentric sphere of radius r ($r < R$) with centre at O. This is the Gaussian surface.

Total charge enclosed by the Gaussian surface

$$q' = \frac{4}{3}\pi r^3 \rho$$

$$= \frac{4}{3}\pi r^3 \frac{q}{\frac{4}{3}\pi R^3} = q\frac{r^3}{R^3}$$

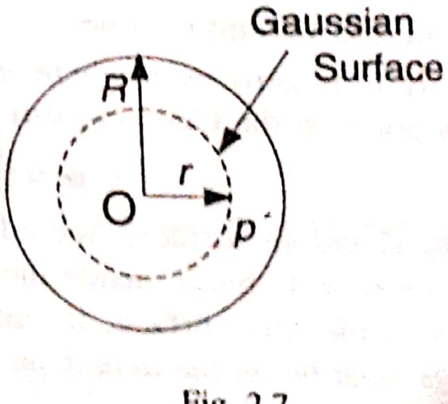


Fig. 2.7

Here, ρ = *charge density* = chargeper unit volume $= \dfrac{q}{\frac{4}{3}\pi R^3}$

The outward flux through the surface of the sphere of radius r is

$$\oint \mathbf{E} \cdot d\mathbf{S} = E(4\pi r^2)$$

Applying Gauss' law,

$$E(4\pi r^2) = \frac{q'}{\varepsilon_0} = \frac{q}{\varepsilon_0}\frac{r^3}{R^3}$$

$$\therefore \qquad E = \frac{1}{4\pi\varepsilon_0}\frac{qr}{R^3},$$

Thus $E \propto r$. At the centre of the sphere, $E = 0$.

Fig. 2.8 shows the variation of electric field E as a function of radial distance r.

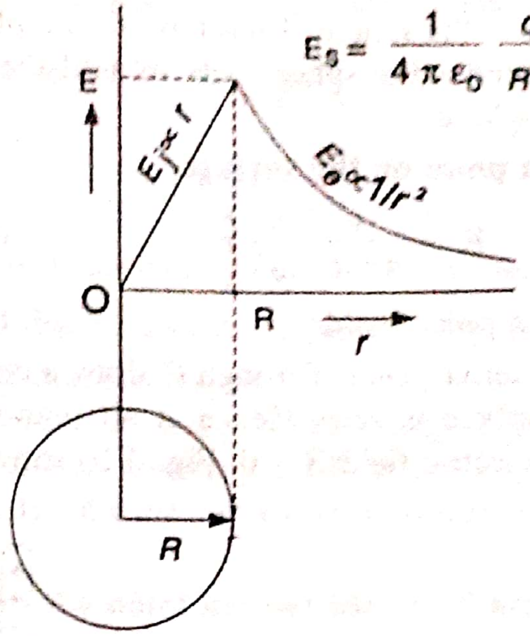


Fig. 2.8

2.6. Electric field due to an isolated Uniformly Charged Conducting Sphere

In an isolated *charged spherical conductor* any excess charge on it is distributed uniformly over its outer surface and there is no charge inside it. *Hence this problem is the same as that of a charged spherical shell.*

Case (i). At an external point

Consider a point P near but outside a uniformly charged sphere of radius R with a charge q (Fig. 2.9). Let σ be the surface density of charge. Then $\sigma = q/(4\pi R^2)$. P is at a distance r from the centre O. Draw a concentric sphere of radius OP with centre O. This is the Gaussian surface. Let E = electric field at any point on this sphere. At every point E is normal to the surface. The flux through this surface is given by

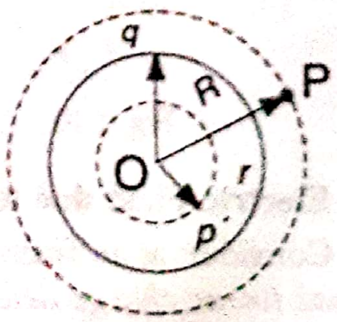


Fig. 2.9

$$\oint \mathbf{E} \cdot d\mathbf{S} = \oint E\,dS = E(4\pi r^2)$$

By Gauss' law. $\qquad E(4\pi r^2) = \frac{q}{\varepsilon_0}$

or $$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

or $$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

The electric field is, therefore, the same as that due to a charge q situated at the centre of the sphere. Therefore, for points outside the sphere, the charges on the conducting sphere behave as if they were concentrated at the centre of the sphere.

Case (ii). At a point on the surface

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{R^2}\hat{r} \qquad (\because r = R)$$

Case (iii). At a point inside

Let P' be an internal point. Through P' draw a concentric sphere. The charge inside this sphere is zero. Hence at all points inside the charged conducting sphere, electric field $\mathbf{E} = 0$. Fig. 2.10 shows the variation of E with r.

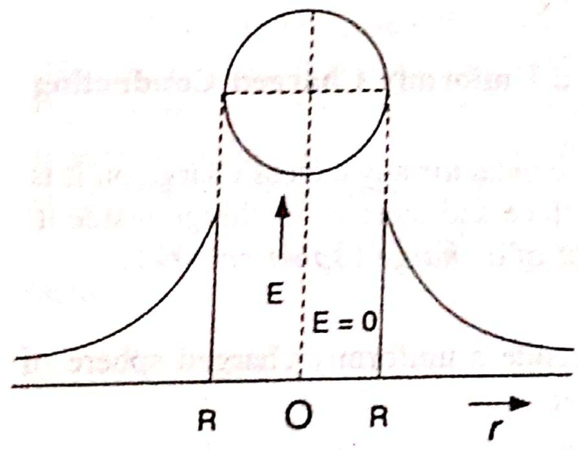


Fig. 2.10

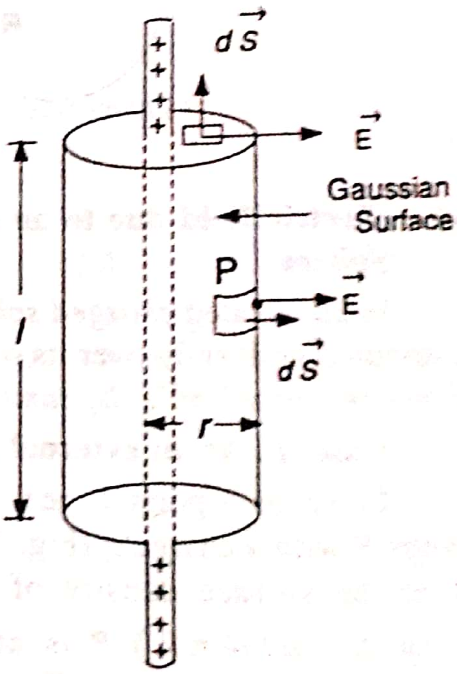


Fig. 2.11

2.7. Electric Field due to an infinite Line of charge

Consider a uniformly charged wire of infinite length having a constant *linear charge density* (charge per unit length) λ. Let P be a point at a distance r from the wire (Fig. 2.11). Let us find an expression for E at P. As a Gaussian surface, we choose a circular cylinder of radius r and length l, closed at each end by plane caps normal to the axis. By symmetry, the magnitude of the electric field E will be the same at all points on the *curved surface* of the cylinder, and directed radially outward. Also E and *dS are along the same direction.*

The electric flux due to curved surface $= \oint \mathbf{E} \cdot d\mathbf{S} = E(2\pi rl)$

The electric flux due to each plane face $= 0$ ($\because$ **E** and $d\mathbf{S}$ are at right angles).

$\therefore$ total flux through the Gaussian surface $= \phi = E(2\pi rl)$

The net charge enclosed by the Gaussian surface $= q = \lambda l$

$\therefore$ by Gauss' law, $E(2\pi rl) = \dfrac{\lambda l}{\varepsilon_0}$

or
$$E = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

The direction of **E** is radially outward for a line of positive charge.

2.8. Electric Field due to a uniform Infinite Cylindrical Charge

Let us consider that electric charge is distributed uniformly within an infinite cylinder of radius R. Let ρ be the charge density (charge per unit volume). We have to find electric field **E** at any point distant r from the axis lying (*i*) inside (*ii*) on the surface (*iii*) outside the cylindrical charge distribution.

Case (i). When the point lies outside the charge distribution

Let P_1 be a point at a distance $r\,(> R)$ from the axis of the cylinder (Fig. 2.12). Draw a coaxial cylinder of radius r and length l such that P_1 lies on the surface of this cylinder. From symmetry, the electric field **E** is everywhere normal to the curved surface and has the same magnitude at all points on it. The electric flux due to plane faces is zero. So the total electric flux is due to the curved surface alone.

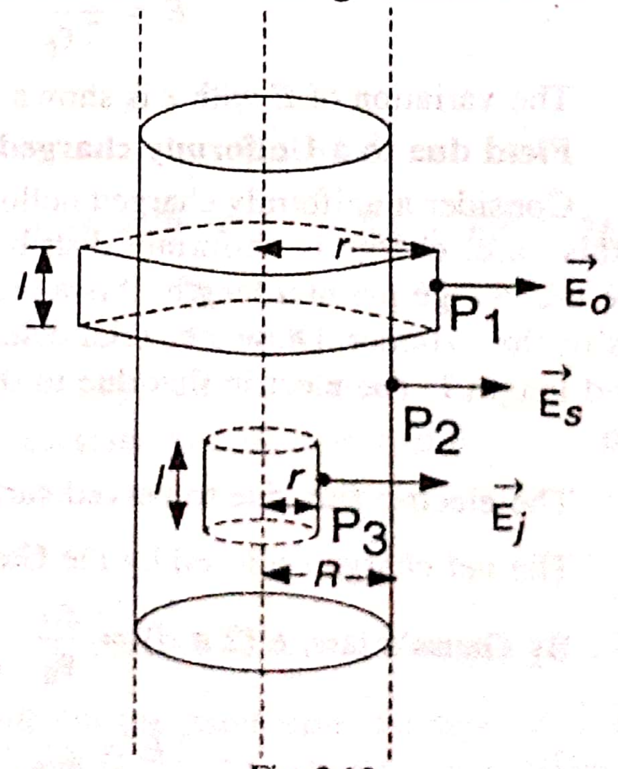


Fig. 2.12

The electric flux due to curved surface

$= \oint \mathbf{E} \cdot d\mathbf{S} = E(2\pi rl)$

The net charge enclosed by the Gaussian surface $= q = (\pi R^2 l)\rho$

$\therefore$ by Gauss' law, $E(2\pi rl) = \pi R^2 l\rho/\varepsilon_0$

or
$$E = \frac{\rho R^2}{2\varepsilon_0 r}$$

Case (ii). When the point lies on the surface of charge distribution (r = R)

Let P_2 be the point on the surface of charge distribution.

By Gauss' law, $E(2\pi Rl) = \pi R^2 l\rho/\varepsilon_0$

$$\therefore \quad E = \frac{R\rho}{2\varepsilon_0}$$

Case (iii). When the point lies inside the charge distribution (r < R)

Let P_3 be the point at a distance r ($< R$) from the axis of the cylinder. Consider a coaxial cylindrical surface of radius r and length l such that P_3 lies on the curved surface of this cylinder.

The charge q' inside this Gaussian surface $= \pi r^2 l\rho$

By Gauss' law, $E(2\pi rl) = \pi r^2 l\rho/\varepsilon_0$

$$\therefore \quad E = \frac{\rho r}{2\varepsilon_0}$$

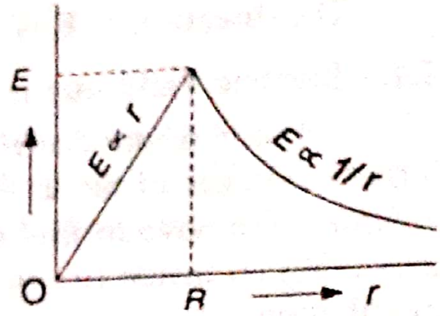


Fig. 2.13

The variation of E with r is shown in Fig. 2.13.

Field due to a Uniformly charged Hollow Cylinder

Consider a uniformly charged hollow cylinder of radius R (Fig. 2.12) In this case, charge is uniformly distributed over cylindrical surface. Let λ be the charge per unit length. P is a point at a distance r ($> R$) from the axis of the cylinder. Draw a coaxial cylindrical Gaussian surface of radius r and length l. The electric flux due to the top and bottom circular caps is zero.

The electric flux due to curved surface $= \oint \mathbf{E}\cdot d\mathbf{S} = E(2\pi rl)$

The net charge enclosed by the Gaussian surface $= q = \lambda l$

By Gauss's law, $E(2\pi rl) = \dfrac{\lambda l}{\varepsilon_0}$

$$\therefore \quad E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

The direction of the electric field is radially outwards.

E is independent of the radius R of the charged cylinder.

Let σ be the surface density of charge on the cylinder. If P is infinitely close to the cylinder, then $\lambda = 2\pi R\sigma$.

$$\therefore \quad E = \sigma/\varepsilon_0$$

If we construct a Gaussian surface inside the hollow cylinder, it will enclose no charge. Therefore, the electric field inside a charged hollow cylinder is zero.

Note. If the cylinder is a conductor, the whole charge is confined to the surface. So the field due to a *charged infinite cylindrical conductor* or a *cable* can be calculated in the same way as discussed in the preceding example.

2.9. Electric Field due to an infinite plane sheet of charge

Consider a thin, *nonconducting*, infinite plane sheet of charge (Fig. 2.14). Let the *surface charge density* (charge per unit surface area) be σ. Let P be a point at a distance r from the sheet. We want to calculate **E** at P. A convenient Gaussian surface is a "pill box" of cross-sectional area A and height $2r$, arranged to pierce the plane as shown. P' is symmetrical with P, on the other side of the sheet. From symmetry, **E** points at right angles to the end caps and away from the plane. Also its magnitude will be same at P and P'.

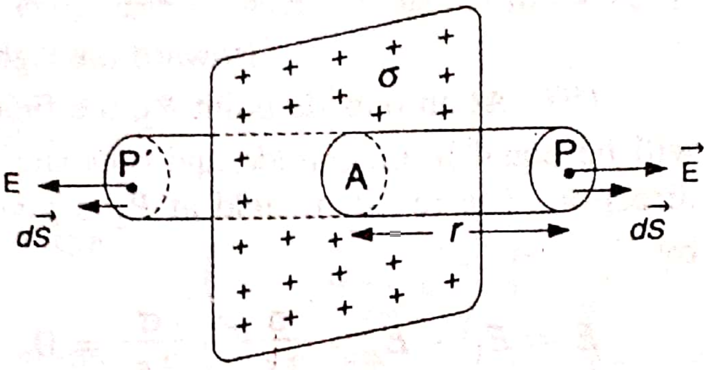


Fig. 2.14

$\therefore$ the flux through the two plane ends is

$$\phi = \oint \mathbf{E} \cdot d\mathbf{S} + \oint \mathbf{E} \cdot d\mathbf{S} = \oint E\, dS + \oint E\, dS = EA + EA = 2EA.$$

The flux through the curved surface of the Gaussian cylinder is zero because **E** and d**S** *are at right angles everywhere on the curved surface.*

$\therefore$ total flux through the Gaussian cylinder $= \phi = 2\,EA$

Net charge enclosed by the Gaussian cylinder $= q = \sigma A$

By Gauss' law, $2\,EA = \dfrac{\sigma A}{\varepsilon_0}$

$$\therefore \qquad E = \frac{\sigma}{2\,\varepsilon_0}$$

E is independent of the distance of the point from the sheet. E is the same for all points on each side of the plane. An infinite sheet of charge cannot exist physically. However this result is true even for sheets of finite sizes if the points chosen are not near the edges and the distance r is small compared to the dimensions of the sheet.

2.10. Field due to Two parallel sheets of charge

Two plane parallel infinite sheets of charge with equal and opposite charge densities $+\sigma$ and $-\sigma$ are shown in Fig. 2.15. The magnitude of

$$\therefore \qquad E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

Let us put a second point charge q' on this surface. The force **F** that acts on this charge q' is

$$\mathbf{F} = q'\,\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{qq'}{r^2}\,\hat{r}$$

The direction of this force is along the direction of **E**, *i.e.*, along the radius vector drawn from q to q'. This is Coulomb's law.

Solved Problems

1. *A small sphere whose mass m is 1×10^{-6} kg carries a charge q of 2×10^{-8} C. It hangs from a silk thread which makes an angle of 30^0 with a large, charged conducting sheet. Calculate the surface charge density σ for the sheet.*

Sol. The electric field near a charged conducting sheet is

$$E = \sigma / \varepsilon_0$$

The electric force acting on the sphere is

$$F = qE = q\sigma / \varepsilon_0$$

The sphere is in equilibrium under the action of three forces:

(*i*) weight of the sphere, mg, acting vertically downwards

(*ii*) the electric force F acting horizontally,

(*iii*) the tension T in the thread acting at an angle θ with the vertical (Fig. 2.21).

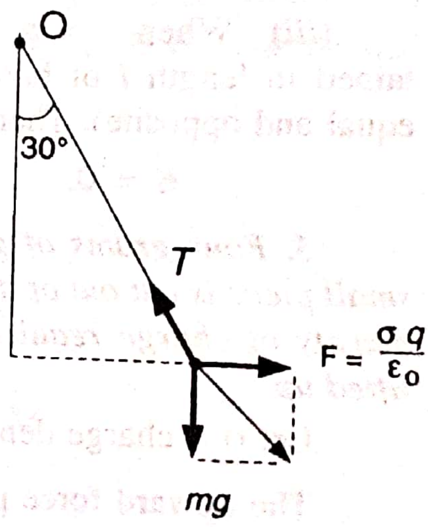


Fig. 2.21

From Fig. 2.21, $\dfrac{F}{mg} = \tan\theta$

$$\frac{q\sigma}{\varepsilon_0\, mg} = \tan\theta$$

$$\therefore \quad \sigma = \frac{\varepsilon_0\, mg \tan\theta}{q} = \frac{(8.85 \times 10^{-12})(1 \times 10^{-6})(9.8)\tan 30^\circ}{2 \times 10^{-8}}$$

or $\quad \sigma = 2.5 \times 10^{-9}\,\text{Cm}^{-2}$.

2. *Two long and coaxial hollow cylinders of radii a and b carry uniformly-distributed equal and opposite charges, with a linear charge density λ. Find the electric field at a point distant r from the axis when (i) r < a, (ii) a < r < b, (iii) r > b (Fig. 2.22).*

Sol. Draw a coaxial Gaussian cylindrical surface of length l and radius r.

The flux through this surface = $\phi = E(2\pi rl)$.

Let q = *net* charge enclosed by the Gaussian surface.

By Gauss's law,

$$E(2\pi r l) = \frac{q}{\varepsilon_0}$$

or $$E = \frac{q}{2\pi\varepsilon_0 rl} \qquad ...(1)$$

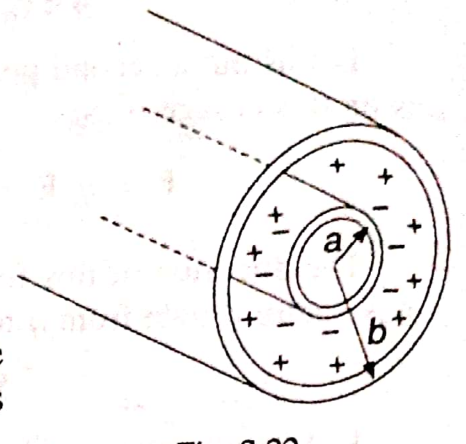


Fig. 2.22

(*i*) When $r < a$, the Gaussian surface is inside the inner cylinder so that it encloses no charge *i.e.*, $q = 0$. Then Eq. (1) gives

$$E = 0$$

(*ii*) When $a < r < b$, the Gaussian surface lies in between the two cylinders. So it encloses the charge contained in a length l of the inner cylinder only *i.e.*, $q = \lambda l$. Then Eq. (1) gives

$$E = \frac{\lambda l}{2\pi\varepsilon_0 rl} = \frac{\lambda}{2\pi\varepsilon_0 r}$$

(*iii*) When $r > b$, the Gaussian surface encloses the charge contained in length l of both the cylinders *i.e.*, $q = 0$ (because charges are equal and opposite). Then Eq. (1) gives

$$E = 0.$$

3. *Four grams of gold are beaten into a thin leaf of area 1 m². A small piece is cut out of it and placed on a conductor. Calculate the surface density of charge required by the conductor so that the gold leaf is just lifted up.*

Let σ = charge density on the surface of the conductor.

The upward force per m² due to charge = $\sigma^2/(2\varepsilon_0)$.

Let m = mass per unit area of the gold foil.

Downward force on the gold foil due to its weight = mg

When the gold foil is just lifted up, $\dfrac{\sigma^2}{2\varepsilon_0} = mg$

$$\therefore \qquad \sigma = \sqrt{2\varepsilon_0 mg} = \sqrt{2\times(8.85\times10^{-12})(4\times10^{-3})9.8}$$

$$= 0.833\times10^{-6}\ \text{Cm}^{-2}.$$

4. *A sphere of diameter 0.05 m is charged to a potential of 1000 volts. Calculate the outward pull per unit area.*

Charge on the sphere is

$$q = 4\pi\varepsilon_0 Vr = 4\pi(8.85\times10^{-12})\,1000\times0.025 = 2.78\times10^{-9}\ \text{C}.$$

$$\text{Surface charge density} = \sigma = \frac{q}{4\pi r^2} = \frac{2.78 \times 10^{-9}}{4\pi (0.025)^2}$$

$$= 3.54 \times 10^{-7}\ \text{Cm}^{-2}.$$

$\therefore$ outward pull per unit area is

$$p = \frac{\sigma^2}{2\varepsilon_0} = \frac{(3.54 \times 10^{-7})^2}{2 \times (8.85 \times 10^{-12})} = 7.08 \times 10^{-3}\ \text{Nm}^{-2}.$$

5. *If the radius and surface tension of a spherical soap bubble be r and T respectively, show that the charge required to double its radius would be* $8\pi r[\varepsilon_0 r(7Pr + 12T)]^{1/2}$ *coulomb where P is the atmospheric pressure.*

Before the bubble is charged, the pressure p_1 inside the bubble is greater than the pressure P (atmospheric) outside by $4T/r$, *i.e.*,

$$p_1 = P + \frac{4T}{r} \qquad ...(1)$$

When the bubble is given a charge q, it experiences outward pull (electrostatic pressure) $\sigma^2/(2\varepsilon_0)$ and expands. If $2r$ is the new radius, then its volume will increase to 8 times. So the pressure inside the bubble will decrease to $p_1/8$. Thus

$$\frac{p_1}{8} + \frac{\sigma^2}{2\varepsilon_0} = P + \frac{4T}{(2r)}$$

or
$$\frac{p_1}{8} = P + \frac{4T}{(2r)} - \frac{\sigma^2}{2\varepsilon_0}$$

But
$$\sigma = \frac{q}{4\pi(2r)^2}.$$

$\therefore$
$$\frac{p_1}{8} = P + \frac{4T}{2r} - \frac{q^2}{512\pi^2 r^4 \varepsilon_0}$$

Substituting the value of p_1 from Eq. (1), we get

$$\frac{P}{8} + \frac{4T}{8r} = P + \frac{4T}{2r} - \frac{q^2}{512\pi^2 r^4 \varepsilon_0}$$

or
$$\frac{q^2}{512\pi^2 r^4 \varepsilon_0} = \frac{7P}{8} + \frac{3T}{2r}$$

or
$$q^2 = 64\pi^2 r^3 \varepsilon_0 (7Pr + 12T)$$

or
$$q = 8\pi r[r\varepsilon_0 (7Pr + 12T)]^{1/2}.$$

3

Electric Potential

3.1. Potential Difference

Consider an isolated point charge $+q$ lying at O (Fig. 3.1). A and B are two points in its electric field. Let W_{AB} be the work done by an external agent in moving a unit positive charge from A to B. *We may define the potential difference between two points in an electric field as the amount of work done in moving a unit positive charge from one point to the other against electrical forces, or* $W_{AB} = V_B - V_A$.

Fig. 3.1

Here, V_A and V_B stand for the potentials at A and B.

The SI unit of potential difference is *Volt*.

The potential difference between two points is 1 volt if 1 joule of work is done in moving 1 coulomb of charge from one point to the other against electrical forces.

Electric Potential. If A is at infinity, then $V_A = 0$.

$$\therefore \qquad W = V_B$$

Here, W is the work done in moving a unit positive charge from infinity to the point B. V_B is the potential at B.

Hence the *electrical potential at a point is defined as the amount of work done in moving a unit positive charge from infinity to that point, without acceleration, against electrical forces.*

The potential at a point near an *isolated positive charge* is *positive*. The potential at a point near an *isolated negative charge* is *negative*.

Equipotential Surface. If all the points of a surface are at the same electric potential, then the surface is called an "equipotential surface".

In the case of an isolated point charge, all points equidistant from the charge are at the same potential. Thus equipotential surfaces in this case will be a series of concentric spheres with the point charge as their centre. The potential will, however, be different for different spheres.

If a test charge q is moved between any two points on an equipotential surface through *any* path, the work done is zero. This is because the potential difference between two points A and B is defined by

$$V_B - V_A = W_{AB}/q.$$

If $V_A = V_B$, then $W_{AB} = 0$. Hence the electric field vector **E** must be everywhere normal to an equipotential surface.

Two equipotential surfaces *cannot* intersect each other.

3.2. Electric Potential as line Integral of Electric Field

Let A and B be two points in a non-uniform electric field (Fig. 3.2). Let an external agent move a test charge q from A to B along *any* path. Let **E** be the electric field at any point P. The electric field exerts a force $q\mathbf{E}$ on the charge q. The external agent must apply a force $\mathbf{F} = -q\mathbf{E}$ in order to move q without acceleration.

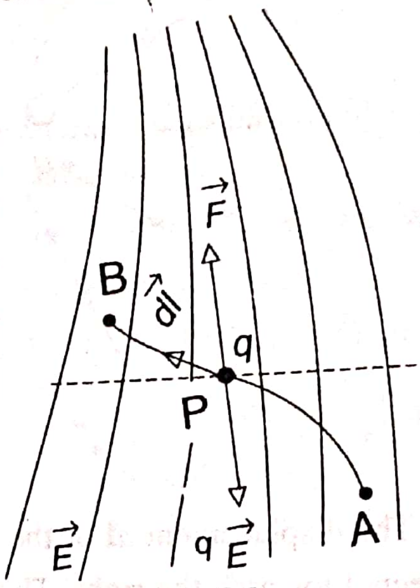


Fig. 3.2

The work done for a small displacement $d\mathbf{l}$ along $AB = \mathbf{F}.d\mathbf{l}$

$\therefore$ total work done in moving the charge from A to B is

$$W_{AB} = \int_A^B \mathbf{F}.d\mathbf{l} = -q\int_A^B \mathbf{E}.d\mathbf{l} \quad (\text{Substituting} \, \mathbf{F} = -q\,\mathbf{E})$$

or $$\frac{W_{AB}}{q} = -\int_A^B \mathbf{E}.d\mathbf{l}.$$

But, by definition, W_{AB}/q is the potential difference $V_B - V_A$ between the points A and B. Thus

$$V_B - V_A = -\int_A^B \mathbf{E}.d\mathbf{l}$$

If the point A lies at infinity, $V_A = 0$. Then the potential at the point B is

$$V_B = -\int_\infty^B \mathbf{E}.d\mathbf{l}$$

3.3. Potential at a point due to a Point Charge

Let $+q$ be an isolated point-charge situated in air. P is a point distant rom $+q$ (Fig. 3.3).

$$\left.\begin{array}{l}\text{The magnitude of the electric field}\\ \text{at a distance } r \text{ from the charge } +q\end{array}\right\} = E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

The potential at P is given by

$$V = -\int_{\infty}^{r} \mathbf{E} \cdot d\mathbf{l} \qquad \ldots (1)$$

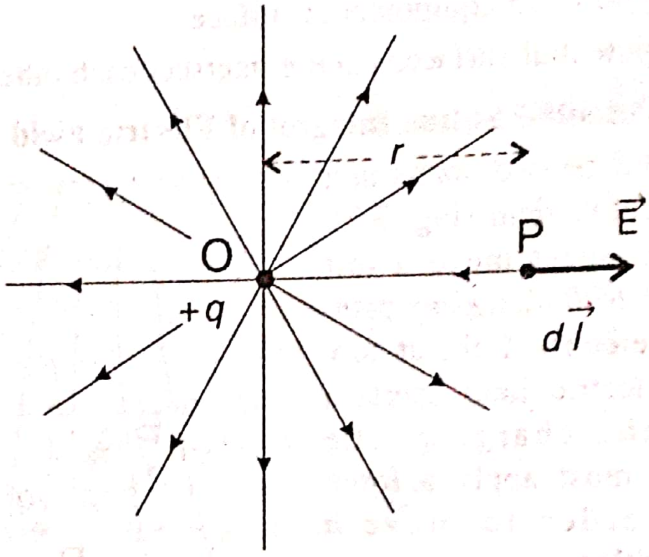


Fig. 3.3

The displacement $d\mathbf{l}$ of the unit charge is directed towards the left. E is directed towards the right. Thus the angle between E and $d\mathbf{l}$ is 180°.

$\therefore \qquad \mathbf{E} \cdot d\mathbf{l} = E\,dl \cos 180^\circ = -E\,dl$

r is measured from the charge $+q$ as origin. As we move a distance $d\mathbf{l}$ to the left, the value of r decreases. Thus $dl = -dr$.

$\therefore \qquad \mathbf{E} \cdot d\mathbf{l} = -E\,dl = E\,dr$

Eq. (1) becomes

$$V = -\int_{\infty}^{r} \mathbf{E} \cdot d\mathbf{l} = -\int_{\infty}^{r} E\,dr = -\frac{q}{4\pi\varepsilon_0}\int_{\infty}^{r} \frac{dr}{r^2}$$

$$\therefore \qquad V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

Potential Difference between two Points. Let r_A and r_B be the distances of two points A and B from the charge $+q$. Potential difference between A and B is given by

$$V_B - V_A = -\frac{q}{4\pi\varepsilon_0}\int_{r_A}^{r_B} \frac{dr}{r^2} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_B} - \frac{1}{r_A}\right)$$

Potential due to a Group of Point Charges. Potential is a *scalar* quantity. In order to find the potential at any point due to a group of point charges, we calculate the potential due to each individual charge, considering the other charges to be absent, and then add up these potentials *algebraically*.

Thus, if there are n point-charges, the potential at a point due to them is

$$V = \frac{1}{4\pi\varepsilon_0} \sum_n \frac{q_n}{r_n},$$

where q_n is the value of the nth charge and r_n is the distance of this charge from the point in question.

If the charge distribution is continuous, then

$$V = \frac{1}{4\pi\varepsilon_0} \int \frac{dq}{r},$$

where dq is a small element of the charge distribution and r is its distance from the point at which V is to be calculated.

Example 1. *Find the potential at the centre of a 1.0 m square having charges q, –2q, 3q and 2q at its corners. ($q = 1.0 \times 10^{-8}$ C).*

Sol. The centre point O is at a distance of $r = 1/\sqrt{2}$ or 0.71 m from each corner (Fig. 3.4).

The potential at O due to all the four charges is

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{r} + \frac{-2q}{r} + \frac{3q}{r} + \frac{2q}{r}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{4q}{r}$$

$$= (9 \times 10^9) \times \frac{4 \times (1 \times 10^{-8})}{0.71}$$

$$= 507 \text{ volt.}$$

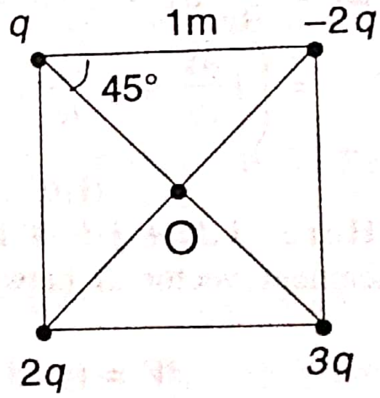


Fig. 3.4

Example 2. *Two charges +q and – 3q are separated by a distance of 1 m. At what points on its axis is the potential zero ?*

Sol. Let the potential be zero at A and B (Fig. 3.5).

(*i*) The point A is on the right of $+q$ at a distance x. Its distance from $-3q$ is $(1 - x)$.

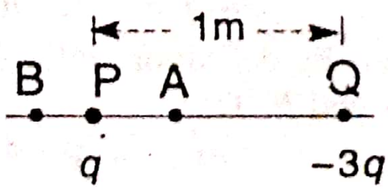


Fig. 3.5

For $V = 0$, we require $\dfrac{q}{4\pi\varepsilon_0 x} + \dfrac{-3q}{4\pi\varepsilon_0(1 - x)} = 0$

$\therefore x = 0.25$m, on the right of $+ q$.

(*ii*) The point B is on the left of $+q$ at a distance x. Its distance from $-3q$ is $1 + x$.

The total potential at B due to $+q$ and $-3q$ is

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{x} - \frac{3q}{1+x}\right)$$

For $V = 0$, we require

$$\frac{q}{x} - \frac{3q}{1+x} = 0$$

$\therefore$ $x = 0.5$ m, on the left of $+q$.

3.4. Relation between Electric Field and Electric Potential

We can calculate the electric field **E** if potential function V is known throughout a certain region of space. Consider two neighbouring points $A\,(x, y, z)$ and $B\,(x + dx,\ y + dy,\ z + dz)$ distance $d\mathbf{l}$ apart in the region (Fig. 3.6). Let the value of potential at A and B be V and $V + dV$ respectively. Let dV be the change in potential in going from A to B. Then

$$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dy + \frac{\partial V}{\partial z}dz$$

$$= \left(\mathbf{i}\frac{\partial V}{\partial x} + \mathbf{j}\frac{\partial V}{\partial y} + \mathbf{k}\frac{\partial V}{\partial z}\right) \cdot (\mathbf{i}\,dx + \mathbf{j}\,dy + \mathbf{k}\,dz)$$

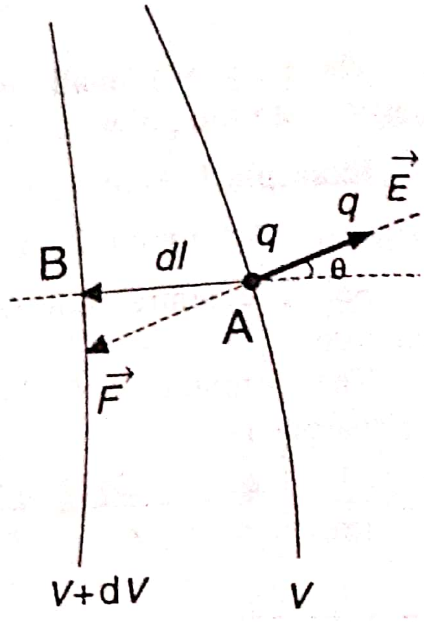


Fig. 3.6

Here, $\mathbf{i}\,dx + \mathbf{j}\,dy + \mathbf{k}\,dz$ is the displacement vector $d\mathbf{l}$ between A and B. Thus

$$dV = (\text{grad } V) \cdot d\mathbf{l} \qquad \text{... (1)}$$

The work done by the external agent in moving a test charge q from A to B along $d\mathbf{l}$ is

$$dW = \mathbf{F} \cdot d\mathbf{l} = -\,q\,\mathbf{E} \cdot d\mathbf{l}.$$

or $\quad dW/q = -\,\mathbf{E} \cdot d\mathbf{l}$

But, by definition, dW/q is the potential difference dV between the points A and B. Thus

$$dV = -\,\mathbf{E} \cdot d\mathbf{l}. \qquad \text{... (2)}$$

Comparing Eqs. (1) and (2),

$$\mathbf{E} = -\text{ grad } V = -\,\nabla V. \qquad \text{... (3)}$$

Thus the electric field at any point is the negative of the gradient of potential at that point. The minus sign indicates that **E** points in the direction of decreasing V.

Let E_x, E_y and E_z be the components of **E** along x, y and z axes. Then

$$E_x = -\frac{\partial V}{\partial x},\quad E_y = -\frac{\partial V}{\partial y},\quad E_z = -\frac{\partial V}{\partial z}.$$

3.5. Potential at a point due to a Uniformly charged Conducting Sphere

When a conducting sphere is given a charge, the charge is distributed uniformly on the surface of the sphere. Let R = radius of the sphere (Fig. 3.7). The sphere is given a charge $+q$.

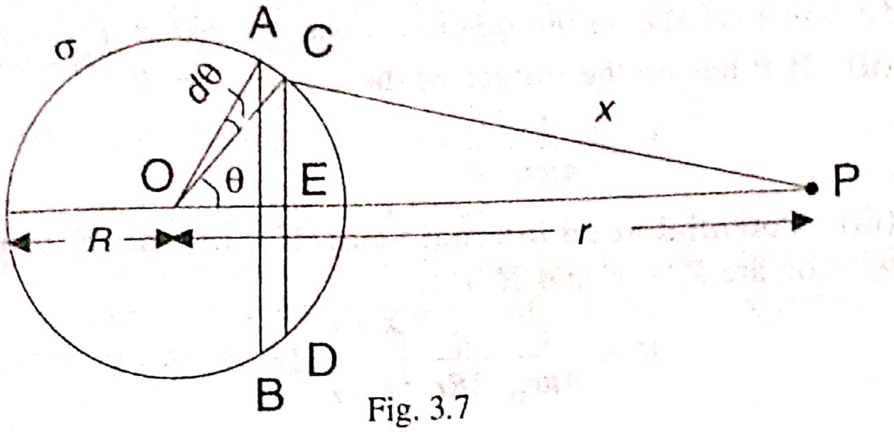


Fig. 3.7

Surface density of charge = $\sigma = q/(4\pi R^2)$

(i) Potential at an external Point. P is a point at a distance r from the centre of the sphere. Draw two parallel planes AB and CED so as to form an annular ring.

$CP = x$, $\angle COP = \theta$, $\angle COA = d\theta$, $CE = R \sin \theta$, $AC = R\, d\theta$

Area of the ring = $(2\pi R \sin \theta)\ (R d\theta) = 2\pi R^2 \sin\theta\, d\theta$

Charge on the ring $ABCD = \sigma\ (2\pi R^2 \sin\theta\, d\theta) = \frac{1}{2} q \sin \theta\, d\theta$

Each point on this narrow ring is at the same distance x from P.

∴ Potential at P due to the charge on the ring

$$dV = \frac{1}{4\pi\varepsilon_0} \frac{\left(\frac{1}{2} q \sin \theta\, d\theta\right)}{x} \quad \text{... (1)}$$

From ΔCOP, $x^2 = R^2 + r^2 - 2Rr \cos \theta$ ... (2)

Here, both x and θ are variables. Differentiating,

$$2x\, dx = 2\, Rr \sin\theta\, d\theta$$

or
$$\sin \theta\, d\, \theta = \frac{x\, dx}{Rr} \quad \text{... (3)}$$

Substituting in Eq. (1), we get

$$dV = \frac{1}{4\pi\varepsilon_0} \frac{q\, dx}{2Rr}$$

The whole sphere is divided into such narrow rings.

∴ Potential at P due to the whole sphere is given by

$$V = \int_{r-R}^{r+R} \frac{1}{4\pi\varepsilon_0} \frac{q\,dx}{2\,Rr} = \frac{1}{4\pi\varepsilon_0} \frac{q}{2Rr} \int_{r-R}^{r+R} dx$$

$$\therefore \quad V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r} \qquad \ldots (4)$$

Hence *the potential at any point outside a charged sphere is the same as if the whole charge on the sphere is concentrated at its centre.*

(ii) If P lies on the surface of the sphere, $r = R$

$$\therefore \quad V = \frac{1}{4\pi\varepsilon_0} \frac{q}{R} \qquad \ldots (5)$$

(iii) Potential at an internal point. If P lies inside, then the limits of integration are $R - r$ and $R + r$.

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{2Rr} \int_{R-r}^{R+r} dx$$

$$\therefore \quad V = \frac{1}{4\pi\varepsilon_0} \frac{q}{R} \qquad \ldots (6)$$

Thus the potential at all points inside a uniformly charged conducting sphere is the same.

The variation of potential inside and outside the sphere is shown in Fig. 3.8.

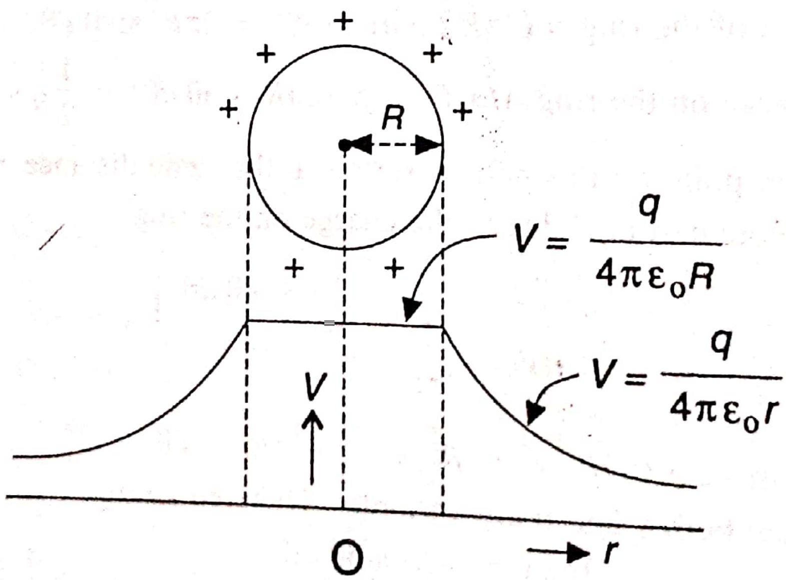


Fig. 3.8

Electric Field. We know that $E = -\,dV/dr$.

$$\left.\begin{array}{l}\text{Electric field outside} \\ \text{the charged sphere}\end{array}\right\} = E_0 = -\frac{d}{dr}\left[\frac{1}{4\pi\varepsilon_0}\frac{q}{r}\right] = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

Its direction is along OP.

$$\left.\begin{array}{l}\text{Electric field inside} \\ \text{the charged sphere}\end{array}\right\} = E_i = -\frac{d}{dr}\left[\frac{1}{4\pi\varepsilon_0}\frac{q}{R}\right] = 0.$$

3.7. Potential and Field due to an Electric Dipole

Fig. 3.11 shows a dipole consisting of two charges $+q$ and $-q$ separated by a distance $2d$.

Dipole moment of the electric dipole $= p = q \times 2d$.

Direction of **p** is from $-q$ to $+q$.

P is a point at a distance r from the centre O.

$\angle POB = \theta$. Let us calculate the potential at P.

AC and BD are drawn perpendicular to OP.

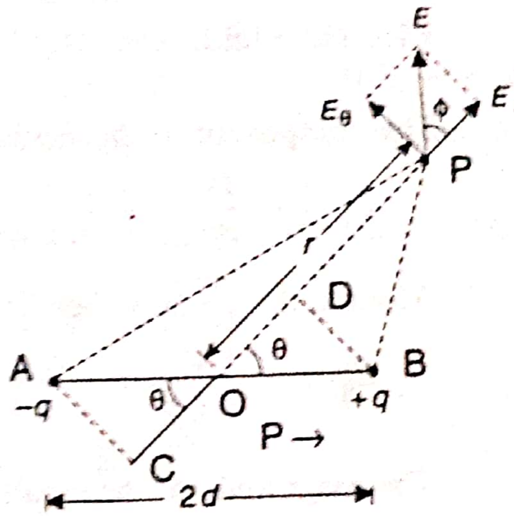


Fig. 3.11

Potential at P due to the charge $+q$ $\Big\} = \dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{q}{PB}\right)$

Potential at P due to the charge $-q$ $\Big\} = \dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{q}{PA}\right)$

Let us assume $r >> d$. Then

$PA = PC = PO + OC = r + d\cos\theta$

$PB = PD = PO - OD = r - d\cos\theta$

The resultant potential at P is

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{(r - d\cos\theta)} - \frac{q}{(r + d\cos\theta)}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{2d\cos\theta}{r^2 - d^2\cos^2\theta}\right)$$

$d^2\cos^2\theta << r^2$ and so it can be neglected. Also, $q \times 2d = p$.

$$\therefore \quad V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

or

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2}$$

Here, $\hat{\mathbf{r}}$ is the unit vector along $\mathbf{r}$. θ is the angle between **p** and **r** (*i.e.*, OP).

Special cases (*i*) When the point P lies on the *axial line* of the dipole on the side of the positive charge, $\theta = 0$ and $\cos\theta = 1$.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$$

(*ii*) When the point P lies on the *axial line* of the dipole on the side of the negative charge, $\theta = 180^\circ$ and $\cos\theta = -1$.

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{-p}{r^2}\right)$$

(*iii*) When the point P lies on the *equatorial line* of the dipole, $\theta = 90^\circ$ and $\cos\theta = 0$. Therefore $V = 0$.

Electric Field. The field at P is calculated using the relation $\mathbf{E} = -\nabla V$.

The component of the field along the radius vector **r** is

$$E_r = -\frac{dV}{dr} = -\frac{d}{dr}\left(\frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}\right) = \frac{1}{4\pi\varepsilon_0}\frac{2p\cos\theta}{r^3}$$

Along a direction perpendicular to **r**, the field component

$$E_\theta = -\frac{1}{r}\frac{dV}{d\theta} = -\frac{1}{r}\left[\frac{d}{d\theta}\left(\frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}\right)\right] = \frac{1}{4\pi\varepsilon_0}\frac{p\sin\theta}{r^3}$$

The magnitude of the resultant field at P is

$$E = \sqrt{(E_r^2 + E_\theta^2)} = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}\left(\sqrt{(3\cos^2\theta + 1)}\right).$$

The angle ϕ which the resultant field **E** makes with **r** is

$$\phi = \tan^{-1}(E_\theta/E_r) = \tan^{-1}\left(\frac{1}{2}\tan\theta\right)$$

Special cases. (*i*) If P is on the axial line, $\theta = 0$.

E is in the direction of **p** and has magnitude $2p/(4\pi\varepsilon_0 r^3)$.

(*ii*) If P is on the equatorial line, $\theta = 90^\circ$.

E is in the direction of $-$**p** and has magnitude $p/(4\pi\varepsilon_0 r^3)$.

3.8. Electric Potential Energy

Definition. *The electric potential energy of a system of fixed point charges is equal to the work that must be done by an external agent to assemble the system, bringing each charge in from an infinite distance.*

Consider a system AB of two charges q_1 and q_2, separated by a distance r (Fig. 3.12). Let us imagine that the charge q_2 at B has been removed to infinity. Now,

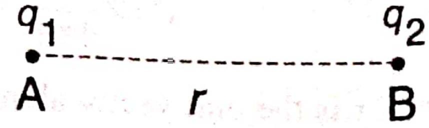


Fig. 3.12

$$\left.\begin{array}{l}\text{the potential at } B \text{ due}\\ \text{to the charge} +q_1\end{array}\right\} = V = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r}$$

This is the work required to be done in order to bring a unit charge from infinity to the point B. Therefore, the work done in bringing the charge q_2 from infinity to the point B is

$$W = q_2 V = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r}$$

This work done represents the electric potential energy (U) of the system of charges q_1 and q_2.

$$\therefore \qquad U = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r}$$

In case of like charges, the potential energy is positive. In case of unlike charges, potential energy is negative.

For systems containing more than two charges, the potential energy of the system is obtained by computing the potential energy for every pair of charges separately and adding the results algebraically.

If we have n different charges in any arrangement in space,

$$U = \frac{1}{2}\left(\frac{1}{4\pi\varepsilon_0}\right) \sum_{i=1}^{n} \sum_{j \neq i} \frac{q_i q_j}{r_{ij}}$$

Example. *An electron (charge $-e$) is placed at each of the eight corners of a cube of side a, and an α-particle (charge $+2e$) at the centre of the cube. Compute the potential energy of the system.*

Sol. The total energy U of the system is the sum of the energies of each pair of charges.

(*i*) There are 12 pairs like A and B with separation a (Fig. 3.13).

(*ii*) There are 12 pairs like A and C with separation $a\sqrt{2}$.

(*iii*) There are 4 pairs like A and G with separation $a\sqrt{3}$.

(*iv*) There are 8 pairs like A and O with separation $a(\sqrt{3}/2)$.

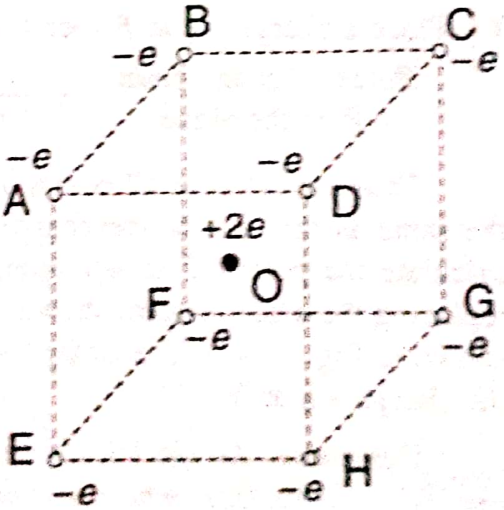


Fig. 3.13

$$U = \frac{1}{4\pi\varepsilon_0}\left[12\frac{(-e)(-e)}{a} + 12\frac{(-e)(-e)}{a\sqrt{2}} + 4\frac{(-e)(-e)}{a\sqrt{3}} + 8\frac{(-e)(2e)}{a(\sqrt{3}/2)}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{e^2}{a}\right)\left[12 + \frac{12}{\sqrt{2}} + \frac{4}{\sqrt{3}} - \frac{32}{\sqrt{3}}\right]$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{e^2}{a}\right)4.32 = (9\times 10^9)\times 4.32\times\frac{e^2}{a}$$

$$= 3.9\times 10^{10}\,\frac{e^2}{a}\text{ J}$$

3.9. Electrical Images

Let XY be an infinite conducting plane which is earthed (Fig. 3.14 a). The plane is at zero potential ($V = 0$). Let a point charge $+q$ be placed at the point A at a distance d from the plane. Due to electrostatic induction, a negative charge is induced on the plane XY. The combined effect of the positive charge at A and the negative charge induced at any point of the conducting plane should make the potential of that point equal to zero.

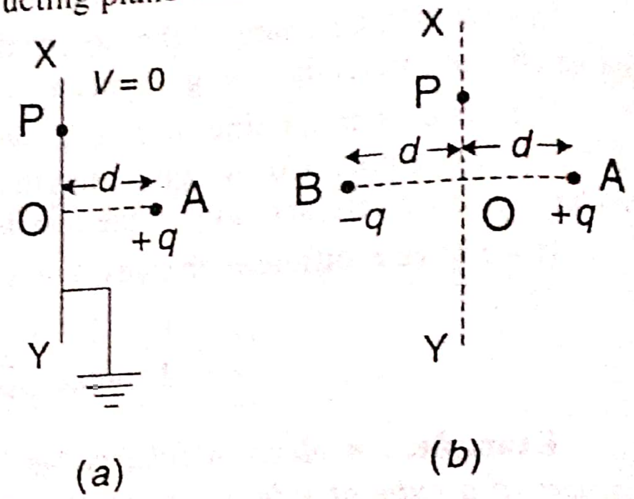


Fig. 3.14

Now, remove the plane XY. Place a charge $-q$ at B such that $AO = BO = d$ (Fig. 3.14 b).

$$\left.\begin{array}{r}\text{Potential at any point}\\ P\text{ on the plane}\end{array}\right\} = \frac{+q}{4\pi\varepsilon_0(AP)} - \frac{q}{4\pi\varepsilon_0(BP)} = 0$$

Thus the effect at all points due to the charge $+q$ at A and $-q$ at B is the same as that due to the charge $+q$ at A and the *plane XY*. So we can calculate the potential at any point in the region S to the right of XY by replacing the actual system shown in Fig. 3.14 a with its equivalent system shown in Fig. 3.14 b. The charge $-q$ at B is called the *electrical image* of the charge $+q$ at A.

Definition. *An electrical image may be defined as a point-charge on one side of a surface, which will produce on the other side of the surface, the same electric field as is produced by the actual electrification of the surface. The actual electrification of the surface is ignored and the image is used instead to find the field.*

Example 1. *A point charge $+q$ is placed in front of an infinite conducting plane connected to earth. Determine by the method of electrical images*

(*i*) *the electric field at a point on the plane*

(*ii*) *the surface density of charge at any point on the plane*

(*iii*) *the force of attraction between the charge and the plane*